Singhai Kshitij Jain

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Summary

AI/ML Engineer, Data Scientist with hands-on experience designing and deploying deep learning and NLP solutions using TensorFlow, PyTorch, and Scikit-learn. Strong foundation in the mathematical principles of machine learning, including linear algebra, probability, and optimization, enabling development of robust prediction models and intelligent automation systems. Experienced in end-to-end solution building-from model development and RLHF-based evaluation to full-stack integration-delivering scalable AI-driven products.

Education

MIT World Peace University — **Pune, Maharashtra**
B.Tech in Computer Science and Engineering — *2022 – 2026*

Higher Secondary — **CBSE Board**
Mathematics — *2022*
Score: 94

Technical Skills

Programming Languages: Python, SQL, JavaScript
WebDev Frameworks: React.js, Node.js, Flask, FastAPI, Express, REST APIs
Databases: MySQL, MongoDB Atlas, Firebase, Pinecone, Redis, FAISS
Machine and Deep Learning: TensorFlow, PyTorch, Scikit-learn, NLP, Time Series Forecasting, Prediction Models, Data Visualization
Generative AI and LLMs: GPT, BERT, LLaMA, LangChain, RAG, Fine-tuning (LoRA, PEFT), Quantization, Speculative Decoding, Teacher-Student Distillation, Local LLMs, Stable Diffusion, GANs
Computer Vision: OpenCV, CNNs, YOLO, Vision Transformers
Cloud and Tools: Linux, AWS, Azure, Git, GitHub, Google Colab, VS Code, Jupyter
Distributed & Network Systems: Distributed computing, TCP/IP, socket programming, API scaling, load-balanced microservices

Projects

AI Agent System (AutoAgent) — *Tech Stack: Python, LLMs, System Automation, GitHub API*
- Built an autonomous AI agent capable of discovering relevant GitHub repositories from user queries, evaluating system compatibility, and executing projects end-to-end.
- Implemented intelligent error handling with iterative reasoning to auto-fix minor issues and generate summaries for complex failures.
- Designed a decision-making pipeline enabling dynamic task selection, execution, and validation without manual intervention.
- **GitHub:** github.com/singhaikshitijjain/AutoAgent

Production AI Deployment Platform (PyForge) — *Tech Stack: FastAPI, Python, Deployment, REST APIs*
- Developed and deployed a production-grade AI system exposing APIs for executing structured Python-based workflows.
- Engineered backend to handle real-world constraints, ensuring stability for production-level code execution.
- Designed system to work specifically with structured inputs, improving reliability and reducing runtime failures.
- **Live Demo:** pyforge.onrender.com

Command-Based ML Pipeline Automation — *Tech Stack: Python, Bash, Scikit-learn, argparse, YAML*
- Architected and built an end-to-end CLI-driven ML pipeline enabling rule-based prediction flows, customizable preprocessing, and automated artifact versioning.
- Implemented multi-model training and comparative evaluation framework, allowing users to edit pipeline configurations dynamically via YAML for flexible experimentation.
- Reduced experiment setup time by 70% while ensuring reproducible and standardized model development workflows.
- **GitHub:** github.com/singhaikshitijjain/mlbuild

Algorithmic Trading Strategy using RL — *Tech Stack: TensorFlow, NumPy, Pandas, RL, LSTM, Electron.js, Matplotlib, Scipy*
- Developed a Reinforcement Learning–based trading agent with 63.8% backtested strategy accuracy.
- Built scalable data pipelines and evaluation modules for systematic backtesting and performance diagnostics.
- Converted the solution into a cross-platform desktop application using Electron.js for interactive strategy execution and visualization.

Experience

AI Data Specialist - Deccan AI Experts — *Intern*
- Conducted RLHF-based evaluation and annotation for both text and image generation models, performing preference ranking, prompt refinement, and multimodal feedback to improve alignment, visual coherence, and factual accuracy.
- Designed and executed multimodal data curation and quality assessment pipelines, defining evaluation rubrics and error taxonomies for image outputs (artifacts, prompt adherence, safety), leading to measurable gains in model reliability and user satisfaction.

Achievements

HackMIT, MIT WPU (ML Bank Fraud Prediction): Lead team and we ranked 3rd among 100+ teams (March 2025)

Publications

Python Package: nececv - Python package reducing text generation time by 40% on low-power devices. (PyPI Link)
Research Paper - Presentation Stage Braincon 2026: AI-Based Hands-Free Operating System Control for Disabled and General Users - integrating computer vision and speech recognition for fully accessible interaction.